Software Project Management
Assignment-0
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which one of the following is not a symptom of the present software crisis?

a. Software is expensive.
b. It takes too long to build a software product.
c. Software is delivered late.
d. Software products are required to perform very complex tasks

Correct Answer: d. Software products are required to perform very complex tasks

Detailed Solution:

Software products are required to perform very complex tasks is not a symptom of the software crisis — it is a root cause. The crisis manifests through high costs, long development times, and late deliveries, which arise because our processes and tools struggle to keep up with ever-increasing system complexity.

QUESTION 2:

Which one of the following is a possible reason for the effort, time, and cost required to develop a program to increase exponentially with the size of the program while using the build and fix style of development?

a. Programming languages lack suitable constructs required for the build and fix style of development
b. Programmers lack competence in effectively using the build and fix style
c. Programmers are able to use abstraction and decomposition techniques in a very limited way while using the build and fix model
d. Programmers are able to make limited use of GO TO statements while using the build and fix model

Correct Answer: c. Programmers are able to use abstraction and decomposition techniques in a very limited way while using the build and fix model

Detailed Solution:

In the build-and-fix approach, there is no systematic design or planning, so as the program grows, the lack of structure and modularity makes understanding, modifying, and debugging increasingly difficult — leading to exponential growth in effort, time, and cost.

QUESTION 3:

A software development company is likely to face which one of the following difficulties, if it adopts the waterfall model in preference to the exploratory (build and fix) program development style in its development projects?

a. Increased development cost
b. Increased development time
c. Decreased program maintainability
d. Increased documentation overhead

Correct Answer: d. Increased documentation overhead

Detailed Solution:

The waterfall model requires extensive documentation at each phase (requirements, design, testing, etc.), whereas the build-and-fix style has little to no formal documentation. While this overhead increases effort, it improves maintainability and reduces long-term costs — unlike the other options, which are generally disadvantages of the build-and-fix approach.

QUESTION 4:

Which one among the following is not a software service type of project?

a. Software maintenance
b. Software customization
c. Outsourced software development
d. Software product development

Correct Answer: d. Software product development

Detailed Solution:

Software product development is not a software service type of project — it is a product-focused effort, where the goal is to build a reusable software product for the mass market or a specific customer. In contrast, software maintenance, customization, and outsourced development are service-oriented activities, where the focus is on providing services for a client's existing or custom systems.

QUESTION 5:

Model building can be considered to be an application of which one of the following techniques?

a. Abstraction
b. Decomposition
c. Aggregation
d. Composition

Correct Answer: a. Abstraction

Detailed Solution:

Model building involves simplifying a complex system by focusing on its essential characteristics while ignoring irrelevant details — which is exactly what abstraction achieves. While decomposition (breaking into smaller parts) and aggregation (grouping) are also useful techniques, the core of model building is creating an abstract representation of the system.

QUESTION 6:

Suppose you are getting an application software developed by a vendor for your organization. Which one of the following characteristics of the software delivered by the vendor to your organization is not a symptom of the present software crisis?

a. Fails to meet user requirements.
b. Expensive.
c. Highly interactive
d. Difficult to alter, debug, and enhance.

Correct Answer: c. Highly interactive

Detailed Solution:

Highly interactive is not a symptom of the software crisis — it is a desirable quality attribute of many modern applications. The crisis is characterized by problems such as failing to meet requirements, high cost, and difficulty in maintenance, whereas high interactivity is a feature that users often demand and appreciate.

QUESTION 7:

A software process model represents which one of the following?

a. The way in which software is developed
b. The way in which software processes data
c. The way in which software is used
d. The way in which software may fail

Correct Answer: a. The way in which software is developed

Detailed Solution:

A software process model (such as Waterfall, Agile, or Spiral) defines the sequence of activities, phases, and workflows used to plan, design, build, test, and deliver a software product — it describes the development lifecycle, not how the software processes data, is used, or may fail.

QUESTION 8:

Which one of the following is not an important difference between the software development projects being undertaken now and the ones that were being undertaken several decades back?

a. Project durations have shrunk from multi-years to few months now
b. Project sizes have increased manifolds
c. Use of high-level languages
d. Increased code and design reuse

Correct Answer: c. Use of high-level languages

Detailed Solution:

Use of high-level languages is not an important difference between software projects now and those several decades ago — high-level languages (e.g., Fortran, COBOL) have been in widespread use since the 1960s. The key differences today include shorter project durations, much larger project sizes, and significantly higher levels of code and design reuse, which have become common only in recent years.

QUESTION 9:

Which one of the following activities spans all stages of a software development life cycle (SDLC)?

a. Coding
b. Testing
c. Project management
d. Design

Correct Answer: c. Project management

Detailed Solution:

Project management is the correct answer, as it spans all stages of the software development life cycle — from inception and planning through development, testing, and deployment to closure. Unlike coding, testing, or design — which are concentrated in specific phases — project management involves continuous planning, monitoring, communication, and risk management throughout the entire project duration.

QUESTION 10:

Which one of the following is the most acceptable definition of a project?

a. A series of activities designed to accomplish some objective
b. An attempt to perform something that has well-defined objectives, and requires both human and other types of resources and usually has to abide by several constraints
c. A coordinated effort to accomplish a well-defined goal but in general can be without any end points
d. Any activity to which human and other resources must be allocated for it to proceed

Correct Answer: b. An attempt to perform something that has well-defined objectives, and requires both human and other types of resources and usually has to abide by several constraints

Detailed Solution:

A data dictionary describes composite data items by breaking them into component elements (b), is commonly used alongside DFDs (d), and includes names and purposes of data items (e). It does not require every data item to map directly to program variables (a), does not represent control flow (c), and does not specify execution sequence (f)—those are implementation and control-flow

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